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AIRCRAFT AND EQUIPMENT MAINTENANCE MANAGEMENT (F-22)

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Chapter 1

MANAGEMENT PHILOSOPHY AND GUIDANCE

1.1. Maintenance Concept.

1.1.1. Maintenance Assistance Requests: advice, assistance, and training pertaining to installation, operation, and maintenance of equipment using currently authorized procedures; authorization for one-time, limited duration or permanent repairs beyond existing technical order (TO) procedures; authorization for limited use of non-listed substitutes to prevent work stoppage. On direction from the Production Superintendent, Action Requests (AR) are submitted through the Field Information Network (FIN) system; then distributed via email through the AR distribution list. Units submit a -107/AR in FIN IAW TO 00-25-107, *Maintenance Assistance* and DAFI 21-103, *Equipment Inventory, Status and Utilization Reporting*.

1.1.2. The Lead Command communicates with the field through Raptor Support Team (RST) Messages. The RST drafts and publishes messages, Integrated Maintenance Information System (IMIS) Event Relationship Files (ERFs), electronic checklist updates, and One Time Inspections (OTI) by direction and under the authority of the Lead Command.

1.1.3. At a minimum, RST messages will be delivered to each F-22 MXG, Quality Assurance (QA) Officer in Charge (OIC)/Superintendent, Product Improvement Manager (PIM), Technical Order Distribution Office (TODO), F-22 Program Office, Air Force Engineering and Technical Service (AFETS), HQ PACAF/A4, NGB/A4, and HQ ACC/A4F22 Weapon System Team (WST) organizational e-mail boxes; distribution to all levels of maintenance is appropriate. Waivers for deferment beyond RST message suspense may be requested under extenuating circumstances through HQ WST. **Note:** Message distribution may require download from a secure site in lieu of e-mail.

1.2. Maintenance Information Systems (MIS).

1.2.1. IMIS is the MIS of input for F-22 aircraft maintenance actions; however, Integrated Maintenance Data System (IMDS) remains the MIS of record and entry for Plans, Scheduling, and Documentation (PS&D) and Maintenance Operations Center (MOC) inputs/actions. IMIS users and managers should consider each MIS applicable when MIS is referenced throughout this and other DAFIs/AFIs/TOs. IMIS will accurately reflect all F-22 maintenance actions. **(T-3)** Personnel documenting maintenance should be trained, qualified, and have access to use IMIS and IMDS as applicable to their Air Force Specialty Code.

1.2.2. Once equipment (e.g., survival kit, ejection seat, gun) is removed from the aircraft through IMIS, subsequent off-equipment maintenance will be documented via the IMIS application Parts Forms. The owning work centers performing off-equipment maintenance require parts manager access to update IMIS with configuration management updates to the end item. **(T-3)**

1.2.3. Electronic aircraft forms (e-forms): The F-22 operates with e-forms and e-TOs that are utilized on remote Portable Maintenance Aids (PMA) that are docked/undocked (synchronized) with a squadron/base level server; this ensures the master aircraft forms and TO database are maintained. Synchronization will be accomplished through a physical

dock/undock process or via a Radio Frequency (RF) link between the PMA and the server where RF infrastructure is available. **(T-3)**

1.2.4. During "comm-out" or RF unavailability conditions, each aircraft will have a dedicated PMA (aircraft e-forms drive) to ensure adequate control of the most current status and maintenance documentation. If in a "comm-out" scenario and a scheduled aircraft is time constrained for dock/undock (e.g., quick turn), the flightline expediter or member of Production will ensure aircraft e-forms, status, and/or debrief actions are reconciled between undocked PMAs and server workstations to facilitate continued sortie generation. **(T-3)**

1.2.5. Only one set of e-forms will be undocked per aircraft from the IMIS server to ensure the integrity of aircraft safety and maintenance documentation. An IMIS hard drive containing all aircraft forms and documentation, commonly referenced as an "All Data Drive," may only be used for review or backup purposes and will be updated in the same fashion as normal aircraft e-forms. The Production Superintendent is responsible for oversight of an "All Data Drive" and anyone operating one must ensure aircraft forms integrity. **(T-3)**

1.2.6. E-forms documentation will not be accomplished on classified hard drives unless pertaining to classified maintenance or with Production Superintendent approval. **(T-3)**

1.2.7. If e-forms documentation is accomplished on a Maintenance Server Workstation (MSW), ensure the aircraft forms drive is docked to the IMIS Server and turned in to support.

1.2.8. The IMIS B+ Technical Order Data (TOD) publication library will be kept current to ensure consistent and timely availability of legacy and support equipment (SE) TOs on the PMA to facilitate maintenance activities. Local TODOs are responsible for ensuring B+ TOD data currency. **(T-3)**

Chapter 2

ROLES AND RESPONSIBILITIES

2.1. Wing Commander (WG/CC) Responsibilities. The WG/CC (or equivalent) will:

2.1.1. Support the F-22 Signature Management Program (SMP) flight test and Radar Cross Section (RCS) Turntable requirements as outlined in **Chapter 11**.

2.1.1.1. Ensure TDY budget includes funding to facilitate deployment requirements for SMP and Radar Cross Section (RCS) test requirements.

2.1.1.2. The minimum requirement for combat-coded aircraft is one aircraft per squadron per year. Any deviation from the minimum number of aircraft required to support SMP will require a formal waiver request from the WG/CC to ACC/A3.

2.1.2. Ensure either Wing and/or Operations Group Commander (OG/CC) attendance at the MXG quarterly Low Observable (LO) Health of Fleet (HoF) briefing.

2.2. Maintenance Group Commander (MXG/CC) Responsibilities. The MXG/CC (or equivalent) will:

2.2.1. Ensure effective engine Total Accumulated Cycle (TAC) utilization to level load Next Scheduled Depot Visit (NSDV) engines to alleviate fleet supportability limiting factors and sustain war ready engines.

2.2.2. Ensure hot wash and lessons learned are documented for each Theater Security Package (TSP) deployment and formal reports are shared with all F-22 units/MAJCOMs. Recommend release within 60 days of return to home station.

2.2.3. Institute a standardized LO long lane REDUX flow plan (See **Attachment 5**) with dedicated LO teams that maximizes efficiency and quality maintenance.

2.2.4. Establish an LO quality assurance program that focuses on aircraft Outer Mold Line (OML) inspections and LO repair processes.

2.2.5. Review Signature Assessment System (SAS) audit trends at least monthly and implement corrective actions when margins exceed established limits (+/- 15%).

2.2.6. Support all directed Radar Cross Section (RCS) test events (e.g., SMP flight test, Radar Cross Section Test Facility (RCSTF), and unit level RF assessments) as outlined in **Chapter 11**.

2.2.7. Provide the WST with tentative aircraft tail numbers for the proposed calendar year SMP and RCSTF dates no later than (NLT) the first day of each fiscal year. If the owning MAJCOM is not ACC, ensure your respective MAJCOM is aware of aircraft movements and schedules for SMP and RCSTF.

2.2.8. The WST is responsible for managing, scheduling, and coordinating SMP and RCSTF events with the applicable OPRs prior to the next year. Additionally, the WST coordinates Air Operations Squadron (AOS)/Coronet movements with HQ ACC AOS/XP for overseas units (3 FW and 154 FW).

2.2.9. Be the approval authority for aircraft to fly more than four sorties in an aero-only configuration.

2.2.10. Establish home-station OML inspection frequency.

2.2.11. Establish a quarterly LO HoF briefing.

2.2.12. Establish a quarterly Avionics HoF briefing.

2.3. Deputy Maintenance Group Commander (MXG/CD) Responsibilities. The MXG/CD (or equivalent) will:

2.3.1. Review and monitor engine Total Accumulated Cycles (TAC) utilization to level load NSDV engines to alleviate fleet supportability limiting factors and sustaining war-ready engines.

2.3.2. Assign a representative to collaborate as needed with the WST on operations, trends, coordination for program requests, and as a communication interface with external organizations.

2.3.3. Assign a software administrator with DoD 8750 privileges to manage the Advanced Diagnostic Integration Tool (ADIT), Multiplex Bus Fault Isolation (MBFI) systems, and Common Organizational Level Tester (COLT).

2.3.4. Ensure the administrator maintains copies of Authority to Operate documents for ADIT, MBFI, IMIS, and COLT. Maintain Computer Program Identification Numbers (CPIN) accounts for operating system downloads.

Chapter 3

AIRCRAFT MAINTENANCE SQUADRON (AMXS) / FIGHTER GENERATION SQUADRON (FGS) RESPONSIBILITIES

3.1. Production Superintendent.

3.1.1. Exceptional Release (ER) authorities will complete the ER using the most current IMIS aircraft forms data. **(T-3)**

3.1.2. Discrepancies not driving combat status will have a detailed correction action plan in the JCN or scheduled maintenance (i.e., deferred discrepancies, open writeups after five subsequent sorties). **(T-3)**

3.1.3. Establish a delayed discrepancy program for discrepancies to reduce the amount of backlog and improve fleet health. **(T-3)**

3.1.4. Coordinate with RST to facilitate Automated Backup Oxygen System (ABOS) load leveling requests. **(T-3)**

3.1.5. Ensure Hanger Queen management includes aircraft temporary storage <90 days and aircraft short/long term storage requirements per the Interactive Electronic Technical Manual (IETM) and TO 1F-22A-6, *Aircraft Scheduled Inspection and Maintenance Requirements*.

3.2. Flightline Expediter.

3.2.1. The Flightline Expediter will inform MOC and Debrief Section of all ground aborts and spare aircraft usage. When a spare is used, ensure the delivery of the micro-Data Transfer Card (*u*DTC) of the aborted aircraft to the debrief section. **(T-3)**

3.2.2. Inflight Operational Checks (IFOC) are managed by the Flightline Expediter.

3.2.3. IFOCs will only be used for operational checks that cannot be completed on the ground.

3.3. Aircraft Section.

3.3.1. Ensure APG personnel involved with OML inspections complete annual OML Damage Identification and Reporting training (Course Code 22183). **(T-3)**

3.3.2. Validate and reconcile data integrity for APU, Turbine Power Module (TPM), and Air Recharge Compressors (ARC) during aircraft document reviews, to include a review of Automated History Event (AHE) and Type Maintenance Interval data integrity with PS&D**. (T-3)**

3.4. Specialist Section.

3.4.1. Ensure flight line specialists involved with OML inspections complete annual OML Damage Identification and Reporting Training (Course Code 22183). **(T-3)**

3.4.2. While avionics specialists are required to perform PACER WARE, SERENE BYTE messaging, or TCTO reprogramming of avionic systems IAW DAFI 21-101, this function is performed by Joint Mission Planning System (JMPS) for the F-22 platform and is generally transparent to maintainers. Actual on-aircraft reprogramming is automatically accomplished when the newly programmed *u*DTC is installed and systems are powered on by aircrew.

3.4.3. Verify currency of aircraft and MIS software configuration management to include aircraft Operational Flight Program (OFP), default mission loads, ERF, and E-checklist. **(T-3)**

3.4.4. Coordinate with JMPS personnel and IMIS administrators to ensure accurate default mission load data is loaded to the IMIS server to include deployed locations for mass memory updates. Default mission data files must include: 469 GPS/Inertial Navigation System (GINS) waypoint, 406 GINS almanac and 420 GPS maintenance keys. **(T-3)** Updates must be loaded to affected aircraft before commencement of flying. Aircraft returning to home station must have default mission load data updated within 30 days. **(T-3)**

3.4.5. Track the number of days each assigned aircraft goes without power applied to the Communication/Navigation and Identification (CNI) system. **(T-3)**

3.4.6. Ensure Multifunctional Information Distribution System (MIDS) batteries do not exceed the time change requirement to prevent unnecessary key loss. **(T-3)**

3.4.7. Establish and maintain an ABOS program. **(T-3)**

3.4.7.1. Maintain equipment accounts and oxygen replenishment contracts for ABOS servicing as necessary. **(T-3)**

3.4.7.2. Conduct servicing cart scheduled maintenance requirements as defined in TO 35D3-6-46-1, *Operation and Maintenance Instructions with Illustrated Parts Breakdown— Trailer, Oxygen Servicing, Type AF/M32R-3,* and TO 35-1-236WC-1, *Periodic Inspection, Non-Powered Aerospace Ground Equipment, Gas Generating and Dispensing Systems*, located on IMIS B+ TOD viewer. **(T-3)**

3.4.7.3. Coordinate with PS&D to resolve ABOS bottle configuration management discrepancies. **(T-3)**

3.4.7.4. Coordinate with Production section to facilitate ABOS load leveling requests. **(T-3)**

3.4.7.5. Validate and reconcile data integrity for date of manufacture for ABOS bottles, aircraft batteries, and fire bottle squibs during aircraft document reviews. **(T-3)**

3.4.8. At a minimum, appoint one Communications Security (COMSEC) Responsibility Officer (CRO) per unit. **(T-3)**

3.4.9. CROs will maintain and account for all installed cryptographic modules (KOVs) and coordinate with the base COMSEC office for timely turn in of suspected or confirmed bad assets. **(T-3)**

Chapter 4

AIRCREW AND MAINTENANCE DEBRIEF SECTION

4.1. Aircraft Sortie Debrief Requirements.

4.1.1. The primary objective for the Debrief section is to process, report, and track presented faults, not to adjudicate the legitimacy of faults. A successful debrief depends on a thorough understanding of this publication, the F-22 Minimum Essential Subsystem Listing (MESL) (**https://usaf.dps.mil/sites/ACC-A4/A4M/A4PM/AFI%2021123%20ACCSUP%20MESL/Forms/AllItems.aspx?viewpath=%2Fsites%2FACC%2DA4%2FA4M%2FA4PM%2FAFI%2021123%20ACCSUP%20MESL%2FForms%2FAllItems%2Easpx&isAscending=true&sortField=LinkFilename&viewid=c12b359b%2De09d%2D4a9e%2D92f7%2D055038417d1c**), the IMIS Help file, and the Fault Report Code (FRC) nuisance listing. The IMIS Help File contains a flow chart and step-by-step debrief guide that will be utilized to standardize the process across the fleet. **(T-3)** A debrief includes a *u*DTC download to IMIS, pilot interview, and Job Control Number (JCN) creation as required. Aircraft sortie times are tracked through *u*DTC download. This time is automatically calculated from Weight-Off-Wheels (takeoff) to Weight-On-Wheels (landing), plus five minutes or engine shutdown, whichever occurs first. In the event an *u*DTC download fails, flight times will be entered manually by debrief personnel and JCNs will be created using the Crash Survivable Memory Unit (CSMU) data. Notify JMPS personnel of failed *u*DTC download to evaluate serviceability of *u*DTC. Include *u*DTC download capture rates to Aircraft Structural Integrity Program (ASIP) manager in accordance with DAFI 63-140_ACCSUP, *Aircraft Structural Integrity Program and Air and Space Equipment Structural Management*. **(T-3)**

4.1.2. Conduct a Debrief at the termination of every sortie or when a sortie is aborted except under the following circumstances:

4.1.2.1. Hot pit operations. A debrief is not required between hot pit sorties unless an aircrew change occurs. **(T-3)**

4.1.2.2. Ground aborted sorties (step to spare). Deliver aborted aircraft *u*DTC to the Debrief section for processing. The pilot stepping to a spare must use a spare/separate *u*DTC to capture accurate fault data for that aircraft. The maintainer who performed the redball maintenance leading up to the abort will convey pilot remarks into JCN discrepancy in IMIS during aborted aircraft debrief action. The aborted aircrew member will confirm/edit comments of the aborted aircraft upon return from the sortie. **(T-3)**

4.1.2.3. During Rapid Crew Swap (RCS) operations, debrief is not required between RCS sorties unless an aircraft lands with a pilot indicated aircraft grounding condition. **(T-3)** If the condition is corrected without maintenance, the aircrew must report to the debrief section and debrief determining comments that define the actions taken to clear the condition. **(T-3)** Example: Environmental Control System (ECS) Integrated Caution and Warning (ICAW) cleared by pilot-performed Initiated Built-in Test (BIT), reset, or restart.

4.1.2.3.1. RCS debriefing process. Debrief personnel will download the *u*DTC from the first sortie pilot and execute the Process Fault Data (PFD) application. PFD will produce applicable pilot questions in IMIS, and the first pilot must completely answer

them as appropriate and record all aircraft issues. The debrief personnel will suspend the session after completing the first sortie by closing the application window without processing the final screen. The second sortie pilot lands and starts the debrief by downloading their DTC. The second pilot will perform a subsequent debrief for the aircraft and answer their PFD questions. Both sorties will have the debrief session completed in flight sequence order. Debrief personnel will present a final screen report and JCNs to the Fighter Generation Squadron, or Aircraft Maintenance Unit Production section. **(T-3)**

4.1.3. Reporting Faults.

4.1.3.1. Create a JCN for all discrepancies reported by a pilot or presented in the final screen of debrief. **(T-3) Note:** The use of informational notes or local tracker spreadsheets do not constitute discrepancy reporting.

4.1.3.2. JCN narratives will include thorough pilot comments, clarification if Code 3 (flyable) or Code 3 (restricted), sortie deviation information, and Repeat/Recur if applicable. **(T-3)**

4.1.3.3. Faults must be debriefed and documented independently with JCNs if not identified automatically by IMIS as pre-existing in aircraft forms. **(T-3)** JCNs may be reviewed by system Subject Matter Experts (SMEs) to determine if faults are the same as pre-existing JCNs. If they are determined as the same fault, then the most recent debriefed JCN should be closed out with direct reference to the pre-existing JCN. Debrief personnel with SME system knowledge may perform JCN correlation review and cross referencing.

4.1.3.3.1. Update existing JCNs with new data observed by aircrew during subsequent sorties. **(T-3)** If a pilot reported discrepancy already exists in aircraft forms, the pilot must confirm the fault is the same as the existing JCN and provide updates or insights that may help in the fault isolation process of that JCN. **(T-3)**

4.1.3.3.2. Adjudication of FRCs from the debrief final screen that contain the words nuisance, no maintenance action, or software anomaly (e.g., NO MX, NMA, SWA) require JCN creation and may be signed off as Code 9 without consulting the nuisance list unless accompanied by pilot observed degradation. **(T-3)**

4.1.3.4. Faults experienced after the participation in Embedded Training (ET) mode are valid faults. Ensure JCN creation criteria is followed. **(T-3)**

4.1.3.5. Ensure discrepancy events created and/or cleared by the pilot have an appropriate corrective action documented. **(T-3) Note:** System resets or avionics restarts are considered appropriate corrective actions; therefore, no further maintenance action is required (See F-22 MESL, Note 11).

4.1.3.6. When utilizing Manual or ADHOC JCN creation, ensure narratives include pilot comments on new capability or OFP being matured in parentheses, sortie deviation information, and repeat/recur information if applicable. **(T-3)**

4.1.3.7. Debrief sections will update the MOC with two LCNs to use for breakouts upon identifying LO coatings as a lead breakout. **(T-3)** The subsystem which is the causal driver of the LO work must be given to MOC to update both timelines for the maintenance driver (secondary breakout) and the LO lead time driver (primary breakout). **(T-3)**

4.1.4. Landing Status and System Capability Codes.

4.1.4.1. Landing status codes (**Table 4.1**) and system capability codes (**Table 4.2**) are not required to be the same. An aircraft can have a landing status code determination of a higher or lower value than the system capability code. Example: Capability Code 9 for CNI Fail that was reset with no additional malfunctions may be debriefed landing status Code 1. Capability Code 3 for a side weapons bay door failure may be debriefed as landing status Code 2 and the MESL would dictate the appropriate aircraft status. The use of **Attachment 7** is key to understanding the tailored consequences of system degrades and failures to make the appropriate determination of aircraft status in conjunction with the reported landing status and capability codes.

4.1.4.2. Aircraft may be debriefed as landing status Code 3 (flyable) or landing status Code 3 (restricted). Use Technical Order 00-20-1, Chapter 4, *Symbols and Their Use* when determining the appropriate symbol to assign during JCN creation. Use the Aerospace Vehicle Maintenance Status Code Flow Chart in **Table 4.3** to help determine the proper aerospace vehicle maintenance and condition status codes to report.

4.1.4.3. Identical FRCs with an open JCN will be debriefed with a landing status of Code 1 and a system capability Code 0. **(T-3)** Pilot comments that are comparable to a previous JCN will be attached to the established JCN and include the sortie number and date. **(T-3)**

4.1.4.4. Informational notes will not be used to report system malfunction(s). Appropriate system capability codes must be used to report all system performance degradation(s). If performance degradation(s) were experienced and subsequently resolved in air or upon recovery with system resets, then system capability Code 9 should be used.

4.1.4.5. Pilot reported discrepancies that were resolved by system reset/avionics restart will be documented with a system capability Code 9 and will not affect landing status. **(T-3)** Discrepancies will be closed out by debrief with “Signed off by Aircrew (SOBAC)” and associated pilot comments annotated in the corrective action (e.g., Pilot Comment: “Code 1 Info note - CNI degrade ICAW asserted during startup cleared with avionics restart” system 48 will be system capability Code 9, and landing Status will remain Code 1), (e.g., Pilot Comment: Code 2 “FLCS advisory cleared with FLCS reset” System 46 will be capability Code 9 and landing status will remain Code 2). **(T-3)**

Table 4.1. Landing Status Codes.

CODE	STATUS
Code 0	Ground Abort
Code 1	Aircraft Mission Capable (MC) with no additional discrepancies
Code 2	Aircraft or system has minor discrepancies but is capable of further mission assignment.
Code 3	Aircraft or system has major discrepancies in mission essential equipment that may require extensive repair or replacement prior to further mission assignment. The discrepancy may not affect safety-of-flight and the aircraft may be Non-Mission Capable (NMC) flyable.
Code 4	Aircraft or system has suspected or known radiological, chemical, or biological contamination.
Code 5	Aircraft or system has suspected or known battle damage.
Note: Debrief will enter code "8" in MIS for aircraft debriefed as code "4" or "5". MESL or MDS equivalent requirements determine if aircraft status is NMC or Partially Mission Capable (PMC).	

Table 4.2. System Capability Codes.

CODE	STATUS
Code 0	System flown with a known discrepancy; no additional discrepancies noted. System can be used.
Code 1	System used and performed satisfactorily. No maintenance required.
Code 2	System used and performed satisfactorily. A minor malfunction exists, but system is capable of further mission assignment. (e.g., RW DEGD or Pilot Comment)
Code 3	System performance was unsatisfactory. This system did not cause an abort. (e.g., RW DEGD or FAIL)
Code 4	System performance was unsatisfactory. This system caused or contributed to an abort.
Code 5	System out-of-commission prior to takeoff.
Code 6	System installed but not used.
Code 7	System not installed.
Code 8	Aircraft or system has suspected or known radiological/biological contamination.
Code 9	System performance degraded or experienced a minor malfunction that corrected in-flight or post-flight to the satisfaction of the pilot.

Table 4.3. Aerospace Vehicle Maintenance Status Code Flow Chart.

QUESTION	RESPONSE	ACTION
A. Is the aircraft RESTRICTED from use or FLYABLE?	RESTRICTED	NMC (Restricted – **Note 1**)
	FLYABLE	Go to question B
B. Does a discrepancy exist against any system/subsystem/component listed on the FSL that limits or prevents full mission performance?	YES	Go to question C
	NO	FMC
C. Is the system/subsystem/component identified on any BSLs?	YES	Go to question D
	NO	PMC
D. Is the system/subsystem/component identified on all BSLs?	YES	Go to question E
	NO	PMC
E. Is the system/subsystem/component completely inoperative or display degraded performance? (**Note 3**)	INOP	NMC (Flyable – **Note 2**)
	DEGRADED	Go to question F
F. Can the system/subsystem/component still perform at least one wartime/training/test mission?	YES	PMC
	NO	NMC (Flyable - **Note 2**) training/test mission

Notes:

1. Input maintenance status code NMCM, B or S and condition status code A through E as appropriate into the applicable maintenance information system.

2. Input maintenance status code NMCM, B or S and condition status code K, L, M, N, or P as appropriate into the applicable maintenance information system.

3. Degraded systems are those systems that are not fully operational, but function well enough to perform at least one assigned mission or part of an assigned mission.

4.1.5. *u*DTC Operation.

4.1.5.1. The MXG/CC and OG/CC may waive the *u*DTC download for multiple sorties such as RCS, hot pits, cross-country or waived sorties. When waived, the same *u*DTC will be used for any subsequent sorties to capture all data from the flying day and will be downloaded at the end of the day. **(T- 3)** A Code 3 by *u*DTC alone involving scenarios using the same *u*DTC, will reflect appropriate status based on pilot assessment at the completion of each sortie. **(T-3)**

4.1.5.2. If IMIS is unable to read the *u*DTC or the aircrew is unable to provide the *u*DTC (i.e., ground abort for cross-country), debrief personnel will coordinate with the Production Superintendent to ensure the data for the CSMU and the comprehensive engine diagnostic unit (CEDU) are downloaded manually. **(T-3)** Accomplish the download using the PMA

no later than the end of the flying day. Utilize the downloaded information to create a JCN that is specific to the FRC via the IMIS function for "Process AT File." **(T-3)**

4.1.6. Docking, Undocking, and PMA-Only Operations.

4.1.6.1. In scenarios where a classified IMIS terminal or PMA are necessary, such as when there are circumstances that will hinder the use of an IMIS workstation (i.e., scheduled power outage), provide an undocked classified hard drive containing the applicable tail number data prior to the flying day. **(T-3)**

4.1.6.2. Notify Production after aircrew debrief is complete on the IMIS MSW so that a forms update of the PMA hard drive may be accomplished. **(T-3)**

4.1.6.3. Forms update via docking is not required between flights if the landing status is Code 1, Code 2, or Code 9. **(T-3)**

4.1.6.4. A forms update is required if a sortie is debriefed landing status Code 3. The Production Superintendent may approve a review of the Code 3 discrepancy on the IMIS workstation if an update or dock/undock is not feasible. Any Code 3 that has associated parametric data must be docked/undocked. **(T-3)**

4.1.6.5. Manually entered JCNs on the undocked PMA must be reconciled with those created out of debrief and cross-referenced to each other. **(T-3)**

4.1.6.6. If IMIS capability is not available to support debrief (i.e., cross-country, demo, evacuation), the aircrew will determine if the aircraft is airworthy. **(T-3)**

4.1.6.7. Prior to an aircraft sortie, debrief personnel will clear all unprocessed failure data by using the Process Failure Data utility on the IMIS workstation. **(T-3)**

4.1.7. Repeats and Recurs.

4.1.7.1. Repeat/Recur discrepancy consideration: nine-digit FRCs are not the sole determining factor for repeat or recur reporting. The same system/subsystem FRCs with similar consequences must be considered in the determination of repeat/recurs. **(T-3)** Example: Airframe Mounted Accessory Drive (AMAD) oil pressure discrepancies can generate either 46XX or 83XX LCNs or both; a supporting pilot comment would drive the repeat or recur. Any capability code used during debrief must be considered for repeat/recur reporting, debrief personnel will ensure repeat/recurs are identified in the discrepancy narrative in IMIS. **(T-3)**

4.1.7.2. Debrief personnel will assess pilot comments and associated FRCs that constitute a repeat or a recur discrepancy. **(T-3)** Servicing FRCs will not be counted as a repeat/recur and will be cleared out of IMDS (screen 169) by debrief. **(T-3)**

4.1.7.3. Verify data accuracy and reconcile discrepancies with PS&D, to include resolution of uncompleted sortie data upon completion of the flying day. **(T-3)**

4.1.8. Debrief Deficiency Reporting.

4.1.8.1. Debrief personnel are responsible for creating and tracking Diagnostic Health Management Deficiency reports (DHM DRs). **(T-3)**

4.1.8.2. If BIT-generated or manual FRCs are not available and ADHOC FRCs are used, a DHM DR will be generated through the PIM to address the specific DHM/debrief filter deficiency. **(T-3)**

4.1.8.3. Generate DHM DRs when pilots identify problems with debriefing questions. **(T-3)**

4.1.8.4. When nuisance FRCs are debriefed and correlated to valid pilot-reported discrepancies, a DHM DR will be generated to capture the deficiencies. **(T-3)**

4.1.8.5. If FRCs are presented on the final screen, but pilot comments contradict validity of FRC presentation and FRCs are possibly erroneous assertions, generate DHM DRs for possible DHM filtering updates. **(T-3)**

4.1.8.6. A new DR is not required if a master DR already exists. Submit updates to existing master DRs with new/additional information. **(T-3)**

4.1.8.7. Debrief will maintain a log of DHM DRs; the log will contain aircraft sortie date, sortie number, pilot name, pilot comments, and/or FRC details and JDRS submission date**.** **(T-3)**

4.1.8.8. Generate a DHM DR when correlation or sympathetic filtering deficiencies are discovered. **(T-3)** Include the presentation of several FRCs from the same system/subsystem for the same reported failure. **(T-3)** Ensure specific sortie data is included to support engineering analysis (i.e., multiple radar drain module FRCs for one radar high power failure). **(T-3)**

4.1.8.9. Rapid Agile Capabilities Release (RACR) operational test feedback and improvement. CAF units have an acknowledged and coordinated responsibility to inform the WST on emergent modernization issues. 59th TES MxOT are the recipients of this information through both traditional DR reporting and regular coordination with the WST. **(T-2)**

Chapter 5

MAINTENANCE SQUADRON (MXS) RESPONSIBILITIES

5.1. Accessories Flight. The Emergency Oxygen System (EOS) will be maintained by Egress personnel. **(T-3)**

5.2. Fabrication Flight.

5.2.1. Establish a dedicated Low Observables (LO) Outer Mold Line (OML) inspection team with a minimum of two qualified LO personnel for each assigned Aircraft Maintenance Unit (AMU)/Fighter Generation Squadron (FGS). **(T-3)** Ensure any rotation of the OML Inspection Team always allows for continued consistency and proficiency. **(T-3)**

5.2.2. Establish an LO Fleet Health Manager for each assigned AMU/FGS to analyze, plan, coordinate, schedule, and execute all aspects of LO maintenance. **(T-3)** The LO Fleet Health Manager will:

5.2.2.1. Must be a TSgt/MSgt 7-level assigned to this position for a minimum of one year, two years optimal. **(T-3)**

5.2.2.2. Establish, track and report SAS fleet health metrics. **(T-3)**

5.2.2.3. The LO maintenance margin for each aircraft is determined via an OML inspection and use of SAS. Report home station and deployed wing SAS averages to Wing leadership, applicable MAJCOM, and WST each flying day. **(T-3)** Report deployed aircraft separately (e.g., aircraft supporting contingencies or exercises). **(T-3)** Send SAS reports and waterfall charts to the following lead command organizational email boxes: **ACCA4V22@us.af.mil** and **acclgms@us.af.mil**. **(T-3)**

5.2.2.4. Coordinate with Production Superintendents and PS&D to schedule annual OML Audits. **(T-3)**

5.2.2.4.1. Annual OML Audits will be accomplished in accordance with **Attachment 3** and applicable technical guidance. **(T-3)**

5.2.2.4.2. All aircraft scheduled to support a TSP and/or contingency deployment must not have an audit due within 30 days of arriving at the deployed location. **(T-3)**

5.2.2.4.3. Distribute fleet SAS Audit results to the WST and ACC/A4MN LO SME within five days after audit completion using the format provided in **Attachment 3**. **(T-3)** Send SAS Audit reports to the following lead command organizational email boxes: **ACCA4V22@us.af.mil** and **acclgms@us.af.mil**. **(T-3)**

5.2.2.5. Review Radar Cross Section (RCS) test data (i.e., SMP and RCSTF) to assess compliance with established maintenance procedures and implement corrective actions to mitigate quality escapes. **(T-3)**

5.2.3. Designate a Low Observable/Composite Repair Facility (LO/CRF) Manager. The LO/CRF Manager will: **(T-3)**

5.2.3.1. Perform facility inspections to determine repair and maintenance requirements. **(T-3)**

5.2.3.2. Report facility operation deficiencies such as the Heating, Ventilation and Air Conditioning (HVAC) systems, compressed and breathing air systems, electrical systems, and plumbing and drainage systems in the LO/CRF and on assigned real property of the LO/CRF. **(T-3)**

5.2.3.3. Perform tasks related to the overall management and operations of the LO/CRF, including energy management and equipment inventory. **(T-3)**

5.2.3.4. Working in coordination with the appropriate Security Manager(s) and/or Security Offices, ensure all measures are taken to maintain security accreditation of facility as required. **(T-3)**

5.2.3.5. Submit facility work orders through Civil Engineering (CE) customer service. In cases of established/approved contract maintenance for facilities, contact contractor for emergency or out of cycle maintenance. This includes warranty repairs and maintenance required. **(T-3)**

5.2.3.6. Track the completion of contractor and CE work orders. Maintain record of all work to include response time and time required until satisfactory completion of work. **(T-3)**

5.2.3.7. Perform safety inspections as required by technical orders and Air Force Instructions. **(T-3)**

5.2.3.8. Perform escort duties or provide escorts as needed for contract maintenance personnel within special access areas of the LO/CRF. **(T-3)**

5.2.4. The recommended fleet SAS margin average for combat coded aircraft during non-contingency operations is ≤ 60%. This SAS margin average is based on ACC/A9 analysis of the requirements to meet Deployment Operational Capability (DOC) timing and to enable sustained combat operations. A fleet margin average ≤ 60% has also proven to be effective in sustaining a balanced scheduling battle rhythm during daily flying operations.

5.2.5. Low Observable Maintenance Scheduling.

5.2.5.1. The LO fleet health manager coordinates with the AMU/FGS Production Superintendents and PS&D to determine aircraft LO REDUX scheduling priority and timeline. **(T-3)**

5.2.5.2. LO maintenance scheduling is determined by the aircraft SAS margin and specific damage type accumulation **(**Attachments **5 and 6)**. **(T-3)**

5.2.5.3. A combat coded aircraft should be scheduled for a long lane LO REDUX when the SAS margin is > 80% and creep/minor damages are the leading driver.

5.2.5.4. Minor/creep damage (2% or less) accumulation/flight hour is used to predict and schedule a long-range LO REDUX.

5.2.5.5. An LO maintenance flow plan **(Attachment 5)** must be established and followed to ensure aircraft maintenance is completed on schedule. **(T-3)**

5.2.5.6. Moderate damages (2% - 19.9%) are typically repaired during opportunistic maintenance.

5.2.5.7. Spike damages (20% or greater) require immediate repair during the next down time. **(T-3)** Every effort should be made to accelerate repairs to minimize down time.

5.2.5.8. Units may define spikes as >10% if necessary to control growth.

5.2.5.9. A short lane LO REDUX should be scheduled when an aircraft SAS margin is primarily driven by moderate and spike damages that will affect mission readiness and/or to control fleet SAS margin growth.

5.2.5.10. Hangar doors will not be opened during LO maintenance if it will affect environmental, safety, and health controls and/or LO repair cure times. **(T-3)**

5.2.5.11. Booted fastener replacement conditions:

5.2.5.11.1. LO will track and provide a list of aircraft tail numbers with fasteners missing boot material at each AMU/FGS scheduling meeting. **(T-3)**

5.2.5.11.2. AMU/FGS will make it a priority to replace as many booted fasteners as possible from the aircraft list. **(T-3)**

5.2.6. Repair Verification Radar (RVR).

5.2.6.1. The RVR is maintained as special test equipment. This equipment is used by qualified Field Service Engineers (FSE), AFETS personnel, and at the Units discretion, skilled LO Technicians (7-lvl) are allowed to use and perform RVR checks and analysis.

5.2.6.2. The LO Integrated Product Team (IPT) will direct an RVR assessment for non-standard repairs and/or defects not characterized in SAS. **(T-3)**

5.2.6.3. RVR of the Integrated Forebody (IFB) tip is required when moisture intrusion is suspected and/or if nose cap/rain erosion coating damage exposes the composite substrate. **(T-3)**

5.2.6.4. The unit may direct RVR assessments to verify LO repair TOD compliance. This should be limited to minimize flying mission impacts. **Note:** RVR will not be used to independently replace SAS damage/repair values.

5.2.7. Radar Cross Section Test Facility (RCSTF) Turntable.

5.2.7.1. Aircraft undergoing depot maintenance at OO-ALC will be sampled at a rate of one out of every three aircraft. All combat coded aircraft undergoing modification at the Marietta Speedline will be tested. Aircraft tested pre-depot do not require a post-depot measurement. **(T-3)**

5.2.7.2. Aircraft subject to RCSTF testing will be SAS compliant unless directed otherwise by the WST. **(T-3)**

5.2.8. ACC funds six RCSTF drop-in slots per year, including training and test-coded aircraft, to provide additional opportunities for the field to assess their LO maintenance/sustainment efforts and to supplement SMP.

5.2.9. Signature Management Program (SMP) Radar Cross Section Flight Test.

5.2.9.1. The minimum requirement for combat coded aircraft is one aircraft per squadron per year. Reference **Chapter 11** below for additional information and requirements.

5.2.9.2. Aircraft are scheduled to deploy to Edwards AFB for 16 days.

5.2.9.3. Aircraft must be SAS and Engine SAS (ESAS) compliant throughout the entire flight test event. **(T-3)**

5.2.10. Additional RCS testing may be required to obtain traceability of significant Radar Cross Section growth in a critical sector, to include flight test on the range, RCSTF, or RVR. LO engineering coordination with the applicable MAJCOM and the WST will determine the method of testing when the radar cross section status of an aircraft is unknown. **(T-3)**

5.3. Propulsion Flight.

5.3.1. Utilize visual aids (i.e., engine Time Distribution Index (TDI) charts) to ensure a smooth flow of scheduled engine shop visits to level load maintenance and balance flight workload capacity and capabilities. **(T-3)**

5.3.2. Attend, or ensure a designated representative attends the weekly and monthly MXG maintenance planning meetings. **(T-3)**

Chapter 6

MAINTENANCE OPERATIONS RESPONSIBILITIES

6.1. Maintenance Management Analysis (MMA).

6.1.1. Develop guidance to document procedures on IMIS to IMDS transaction management, data mismatch correction and error prevention. As a minimum, guidance will address the following: **(T-3)**

6.1.2. Transaction priorities and acceptable timelines for error correction. **(T-3)**

6.1.3. Reconciliation of interface data prior to equipment transfer. **(T-3)**

6.1.4. Identification and correction of inspection and time change mismatches between IMIS and IMDS. **(T-3)**

6.1.5. Configuration resync of inspection and time change components. **(T-3)**

6.1.6. Utilization reporting for accurate life cycle and flying hour management. **(T-3)**

6.2. Maintenance Operations Center (MOC).

6.2.1. Reported aircraft status must include a detailed breakout; avoid two-digit LCN breakouts. Ensure the pacing JCN is kept up to date with appropriate maintenance actions and primary/secondary LCNs. **(T-3)**

6.2.2. Aircraft status delay codes will be used when the 8-hour fix rate is exceeded. **(T-3)**

6.2.3. Update IMDS Geolocation Code (GEOLOCATE) as applicable. **(T-3)**

6.2.4. When aircraft Purpose Identifier Codes (PIC) are changed to a non-possessed code, MOC will note the Action Request or T.O. 00-25-107 request number as the source for code change. **(T-3)**

Chapter 7

QUALITY ASSURANCE (QA)

7.1. QA Superintendent Responsibilities. Ensure locally established Maintenance Standardization and Evaluation Program (MSEP) assures the integrity of LO maintenance and SAS documentation associated with mission capable status reporting. (T-3) This includes frequent personnel evaluations of aircraft OML inspections, SAS documentation, and LO repair processes IAW established technical order guidance.

7.2. QA Inspector Responsibilities.

7.2.1. Support MIS configuration control. **(T-3)** Configurable support systems that reside in IMIS are necessary to ensure safety of flight for assigned aircraft and sound maintenance practices.

7.2.2. Responsible for contacting the Program Office/CTODCU for verification approval of IETM tasks when tasks are not validated and verified for use within IMIS (i.e., Yes/Yes). **(T-3)** Once verified, QA will submit a Technical Order Data Change Request (TODCR) for IETM correction or a release recommendation making the IETM approved for fleet-wide use. **(T-3)**

7.3. LO QA Inspector Responsibilities.

7.3.1. Review SMP and Turntable results to determine if any LO processes require special attention. **(T-3)**

7.3.2. Ensure a focused effort on process compliance to mitigate quality escapes that could impact RCS.

7.3.3. Assist in developing a highly skilled LO workforce with proactive oversight and guidance.

7.3.4. Update Wing leadership on LO maintenance quality and potential improvement areas during the Health of Fleet (HoF) brief if applicable.

7.4. QA Product Improvement Programs (PIP). F-22 PIM is responsible for the review of TODCR submissions of unverified and recommended changes to maintenance tasks to ensure they are properly and completely documented as required for submission. (T-3) The PIM tracks the TODCR submission through the entire approval process and disposition. (T-3)

Chapter 8

IMPOUND PROCEDURES

8.1. Mandatory Impoundments. F-22 aircraft and/or equipment will be impounded for:

8.1.1. Any uncommanded or inadvertent weapon station extensions. **(T-2)**

8.1.2. Any uncommanded or inadvertent weapons bay door movement. **(T-2)**

8.1.3. Any known contamination such as fuel, hydraulic fluid, Polyalphaolefin (PAO), or oil. **(T-2)**

Chapter 9

ADDITIONAL MAINTENANCE REQUIREMENTS AND PROGRAMS

9.1. Maintenance Recovery Team (MRT).

9.1.1. When establishing dispatch procedures for MRTs, consider the following:

9.1.1.1. Select F-22 divert locations and develop procedures to recover transient or diverted aircraft based on historical divert locations and past MRT dispatches.

9.1.1.2. Strive for minimum MRT footprint. Negotiate for use of host location support equipment at prospective divert or transient locations if applicable.

9.1.1.3. Pilots flying F-22 cross-country sorties must be provided a PMA hard disk drive with aircraft forms, configuration, and support data loaded in the event of unscheduled maintenance requirements. **(T-3)**

9.1.2. When establishing Memorandum of Understanding/Memorandum of Agreement (MOUs/MOAs) between home station and other locations to enable aircraft recovery and documentation of maintenance, consider the following:

9.1.2.1. Establish IMIS reach back to the home station server, by timely transfer of data, and pre-staging opportunities for IMIS data transfer when possible.

9.1.2.2. Develop procedures such that extended MRT operations do not exhaust PMA-only JCN creation capabilities.

9.1.2.3. Explore employment of host location, or geographically near, F-22 Maintainers to serve as, relieve, or augment MRTs.

9.1.2.4. Establish Organizational MOUs/MOAs in this case to enable Maintainers to access the diverted unit's IMIS account for the duration of the MRT. Data cannot cross organizations without data transfer processes occurring between units. Timely pre-coordination between APOs is critical to alleviate security delays.

9.2. Wing Avionics Manager (WAM).

9.2.1. Monitor, data trend, forecast and execute Avionics HoF and Unit-led Avionics maintenance events.

9.2.2. IAW **paragraph 2.2.11**, provide quarterly Avionics HoF with analysis of aircraft statusing performance, systemic challenges, supply system demands, deficiency reporting, and support equipment.

9.2.3. Coordinate with the WST and F-22 PO on deficiency reporting and investigation. The WST will provide the owning MAJCOM weapon system managers a deficiency report and investigation update on a regular basis.

9.2.4. Address issues with avionics testing, support equipment, systems anomalies, and MAJCOM inspection and COMBAT SHIELD evaluation results by leveraging MXG/CC cross tells and F-22 PO Quarterly meeting attendance.

Chapter 10

MAINTENANCE PLANS, SCHEDULING AND DOCUMENTATION (PS&D)

10.1. Engine Management (EM).

10.1.1. Assess engine fleet health through use of visual aids (i.e., engine TDI charts) monthly, at a minimum, during maintenance planning meetings. **(T-3)**

10.1.1.1. Engine TDI Management. Maintain TDI chart capturing TACs remaining on all engines assigned to Stock Record Account Number (SRAN) as outlined in Air Force Tactics, Techniques, And Procedures (AFTTP) *3-3 Aircraft Maintenance* located here: **https://intelshare.intelink.gov/sites/561jts/afttp/aircraft_maintenance/SitePages/Home.aspx**.

10.1.1.2. If IMDS-Configuration Data Base (CDB) and Comprehensive Engine Management System (CEMS) are unavailable for more than 48 hours at home station, or while deployed, all MXG work-centers will use IMIS until IMDS-CDB and CEMS become available. **(T-3)** IMIS has the ability to store approximately a 2-week period of data without IMDS updating, allowing the use of IMIS to continue until IMDS-CDB and CEMS become available.

10.2. Scheduled Maintenance Management.

10.2.1. TCTOs will be reconciled weekly between IMIS and IMDS. **(T-3)** IMDS screen 525 is the system of record for TCTO compliance and fleet management.

10.2.2. Maintenance management will refer to master LCN listing and reconcile statuses weekly with applicable scheduled maintenance LCNs. **(T-3)**

Chapter 11

SIGNATURE MANAGEMENT PROGRAM

11.1. Introduction. CAF 304-83-I/II/IIIA, *F-22 Advanced Tactical Fighter Operational Requirements Document (ORD), Annex B,* states, "A measurement program shall be developed to sample the dynamic RCS upon initial F-22 delivery and throughout the service life of the aircraft. The number of aircraft assessed and allowable variation from established standards should provide 90% confidence that the F-22 RCS requirements are met across the remainder of the F-22 fleet." This ORD requirement is partially fulfilled via annual Signature Management Program (SMP) open air testing. Due to limited range accessibility for the Raptor fleet, fleet size, scheduling, and funding constraints, the F-22 program is not able to measure an adequate number of aircraft via dynamic flight testing. Therefore, RCS testing in the Marietta RCSTF is necessary to fulfil this requirement.

11.2. WST Responsibilities. The WST is responsible for managing, scheduling, and coordinating SMP and RCSTF events with applicable OPRs. Additionally, the WST coordinates AOS/Coronet movement with HQ ACC AOS/XP offices for overseas units (3 FW and 154 FW).

11.3. SMP Open Air Testing.

11.3.1. The F-22 Combined Test Force (CTF) at Edwards AFB will be the responsible test organization (RTO) for all F-22 SMP testing. Capability to execute from other location(s) will be as designated by the RTO. **(T-2)** The 411th Flight Test Squadron (FLTS) is responsible for all test planning, execution, and reporting.

11.3.2. Each combat-coded unit will send a minimum of one aircraft per squadron per year for dynamic flight testing. Any deviation from the minimum number of aircraft required to support SMP will require a formal waiver request from the WG/CC to ACC/A3. **(T-2)**

11.3.2.1. Units should plan to send a minimum of one Crew Chief, one Avionics 7-level technician, and one Low Observable 7-level technician to support the SMP deployment.

11.3.2.2. It is the unit's responsibility to ensure the TDY budget includes funding to facilitate deployment requirements.

11.3.3. Aircraft will be deployed to Edwards AFB for a total of 16 days (aircraft arriving on Thursday and departing on a Friday). The deployment timeline will not be exceeded to satisfy measurement criteria without approval of the owning unit MXG/CC. **(T-3)**

11.3.4. Aircraft must be SAS and ESAS compliant throughout the entire flight test event. **(T-2)** Any aircraft FMC for LO is eligible for deployment consideration. The intent of this effort is to assess fleet signature health based on annual testing of selected unit-owned aircraft, which are representative of "typical" aircraft on the ramp.

11.3.5. The F-22 COMSEC account participating in this test is responsible for notifying the F-22 System Key Controlling Authority (HQ ACC/A3YJ) by message 10 days prior to generation of the Instrumentation System Data Link (ISDL) key. The message will be prepared IAW the COMSEC Call Out format and include participating units, start and end dates, and the justification. The F-22 System Controlling Authority will grant approval upon receipt of the message. The COMSEC account will generate the ISDL key and distribute to the F-22 Secure Interface System (SIS) (i.e., requestor). After receipt of the ISDL key, the F-22 SIS

operator will comply with inventory and accounting requirements outlined in AFMAN 17-1302-O, *Communications Security (COMSEC) Operations,* or locally established guidance for key management. Tests will not be conducted during the fleet-wide annual rekey in September. **(T-2)**

11.3.6. Aircraft in SMP must have a fully functional intra-flight data link (IFDL) with net plan 20 functionality. Net plan 20 will be dedicated to operations involving data transmissions from the IFDL to the ISDL. Mission planning of the IFDL net plans remains the same; however, a change is made to the air vehicle such that selection by the pilot of Net Plan 20 results in the transmission of Time Space Position Instrumentation (TSPI) and aircraft configuration data rather than the current operational data. **(T-2)**

11.3.7. Calendar inspections must not be due within 30 days of aircraft arrival to SMP. **(T-2)**

11.3.8. All maintenance actions will be documented on the PMA only; units must bring classified and unclassified hard drives to support debrief, daily maintenance, and SAS documentation. **(T-2)**

11.3.9. For F-22 SMP reporting, the RTO will be responsible for test planning and test execution. The RTO will generate/release an electronic copy of merged test reports to the WST and LM-Aero Special Technology. In addition, an initial assessment of results will be provided by RTO to HQ ACC/A4F and ACC/A5A. Preliminary finding and results will be communicated to operational units within 30 days. Further analysis will also be provided as detailed in the SMP plan. The F-22 Program Office and the F-22 CTF will formally track performance of measured aircraft and work with LM-Aero Special Technology section to develop metrics on fleet health. **(T-2)**

11.4. Radar Cross Section Test Facility (RCSTF).

11.4.1. Combat-coded aircraft undergoing reversion/depot maintenance at OO-ALC will be sampled at a rate of one out of every three aircraft. All combat-coded aircraft undergoing modification at the Marietta Speedline will be tested. Aircraft tested prior to inducting into depot do not require a post-depot measurement. **(T-2)**

11.4.2. Aircraft will be deployed to Marietta for a total of six days (Thursday through Wednesday). The deployment timeline will not be exceeded to satisfy measurement criteria without approval of the owning unit MXG/CC. **(T-3)**

11.4.3. ACC funds six RCSTF drop-in slots per year to provide additional opportunities for the field to assess their LO maintenance/sustainment efforts and to supplement the SMP. Each combat-code unit will plan to send one aircraft per squadron per year to the RCSTF. **(T-2)**

11.4.3.1. Aircraft subject to RCSTF testing will be SAS compliant unless directed otherwise by the WST. **(T-2)**

11.4.3.2. Units will pick and restore the hoist points of all aircraft undergoing RCS testing via one of the ACC drop-in slots. **(T-2)**

11.4.4. Preliminary findings and results will be communicated to the units within 30 days. **(T-2)**

11.5. LO Repair Verification Radar (RVR).

11.5.1. The F-22 LO RVR is special test equipment (STE) and is used to measure the signature performance of LO repairs and damages within a zonal area. RVR detects non-visual anomalies, ensures repair performance, provides feedback to the maintainer/depot, tracks and monitors recurring damage areas, and defines impacts of unique LO damages not identified in SAS. It will not be used to replace SAS damage and/or repair values. **(T-2)**

JENNIFER HAMMERSTEDT, Brigadier General, USAF
Director of Logistics, Engineering, & Force Protection

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 33-322, *Records Management and Information Governance Program*, 23 March 2020

AFMAN 17-1302-O, *Communications Security (COMSEC) Operations*, 9 April 2021

CAF 304-83-I/II/IIIA, *F-22 Advanced Tactical Fighter Operational Requirements Document (ORD), Annex B,* 17 February 2004

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TO 00-25-107, *Maintenance Assistance*, 1 October 2015

TO 1F-22A-6, *Aircraft Scheduled Inspection and Maintenance Requirements*, 29 April 2021

TO 35-1-236WC-1, *Periodic Inspection, Non-Powered Aerospace Ground Equipment, Gas Generating and Dispensing Systems*, 21 July 2017

TO 35D3-6-46-1, *Operation and Maintenance Instructions with Illustrated Parts Breakdown—Trailer, Oxygen Servicing, Type AF/M32R-3,* 22 July 2017

Prescribed Forms

None

Adopted Forms

DAF Form 847, *Recommendation for Change of Publication*

Abbreviations and Acronyms

ABOS—Automatic Back Up Oxygen System

ADIT—Advanced Diagnostic Integration Tool

AFETS—Air Force Engineering and Technical Service

AHE—Automated History Event

AMAD—Airframe Mounted Accessory Drive

AMU—Aircraft Maintenance Unit

AOS—Air Operations Squadron

APGS—Auxiliary Power Generation System

APO—Advanced Programs Office

APU—Auxiliary Power Unit

AR—Action Requests

ARC—Air Recharge Compressor

ASIP—Aircraft Structural Integrity Program

BIT—Built-In Test

CDA—Commercial Derivative Aircraft

CDB—Configuration Data Base

CEDU—Comprehensive Engine Diagnostic Unit

CEMS—Comprehensive Engine Management System

CNI—Communication/Navigation and Identification

CPIN—Computer Program Identification Number

CRF—Composite Repair Facility

CRO—COMSEC Responsibility Officer

CSMU—Crash Survivable Memory Unit

CTF—Combined Test Force

CTODCU—Central Technical Order Data Control Unit

DBM—Database Manager

DHM—Diagnostic Health Management

DOC—Deployment Operational Capability

DR—Deficiency Report

DTC—Data Transfer Card

ECS—Environmental Control System

EED—Electro Explosive Device

EOS—Emergency Oxygen System

ERF—Event Relationship File

ESAS—Engine Signature Assessment System

FGS—Fighter Generation Squadron

FIN—Field Information Network

FRC—Fault Reporting Code

FSE—Field Service Engineers

GINS—Global Inertial Navigation System

GPS—Global Positioning System

HoF—Health of Fleet

HVAC—Heating, Ventilation and Air Conditioning

ICAW—Integrated Caution and Warning

IDEA—Innovation Development through Employee Awareness

IETM—Interactive Electronic Technical Manual

IFB—Integrated Forebody

IMDS—Integrated Maintenance Data System

IMIS—Integrated Maintenance Information System

IPT—Integrated Product Team

ISDL—Instrumentation System Data Link

JCN—Job Control Number

JDRS—Joint Deficiency Reporting System

JMPS—Joint Mission Planning System

JTRS—Joint Tactical Radio System

LCN—Load Classification Number

LM—Lockheed Martin

LO—Low Observable

MBFI—Multiplex Bus Fault Isolation

MESL—Mission Essential Subsystem Lists

MIDS—Multifunctional Information Distribution System

MIS—Maintenance Information Systems

MMA—Maintenance Management Analysis

MOA—Memorandum of Agreement

MOCC—Maintenance Operations Center Controller

MOU—Memorandum of Understanding

MRT—Maintenance Recovery Team

MSEP—Maintenance Standardization & Evaluation Program

MSW—Maintenance Server Workstation

MXG—Maintenance Group

OFP—Operational Flight Program

OIC—Officer in Charge

OML—Outer Mold Line

PAO—Polyalphaolephin

PIM—Product Improvement Manager

PIP—Product Improvement Programs

PMA—Portable Maintenance Aid

POC—Point of Contact

QA—Quality Assurance

RCS—Rapid Crew Swap

RCS—Radar Cross Section

RCSTF—Radar Cross Section Training Facility

RF—Radio Frequency

RMS—Reliability, Maintainability, and Sustainability

RST—Raptor Support Team

RSVR—Reservoir

RTO—Responsible Test Organization

RVR—Repair Verification Radar

SAS—Signature Assessment System

SIS—Secure Interface System

SIT—System In Test

SOT—Stability Over Time

SME—Subject Matter Expert

SMP—Signature Management Program

SRAN—Stock Record Account Number

STE—Special Test Equipment

TDI—Time Distribution Index

TO—Technical Order

TOD—Technical Order Data

TODCR—Technical Order Data Change Request

TODO—Technical Order Distribution Office

TPM—Turbine Power Module

TSP—Theater Security Package

TSPI—Time Space Position Instrumentation

WST—Weapon System Team

Attachment 2

SAS REPORTING METRICS

A2.1. SAS Reporting Metrics. The LO Maintenance Margin for each aircraft is determined via an OML inspection and use of SAS. Fleet SAS margin numbers must be documented in a format similar to the chart below **(Figure A2.1)**. **(T-3)** Key information includes SAS margin number for each aircraft, wing SAS average and number of aero only panels.

A2.1.1. Properly reported SAS is dependent on all panels being installed on the aircraft and in IMIS. This requires panel specific part number and serial number verification prior to installing a panel on the aircraft. Panels improperly installed in IMIS that do not include part number and serial number will show as uninstalled resulting in the aircraft displaying inaccurate SAS averages. **Note:** There is an unacceptable risk to aircraft radar cross section and aircraft survivability due to substandard maintenance practices or inaccurate maintenance documentation into IMIS and the SAS.

Figure A2.1. SAS Reporting Metrics.

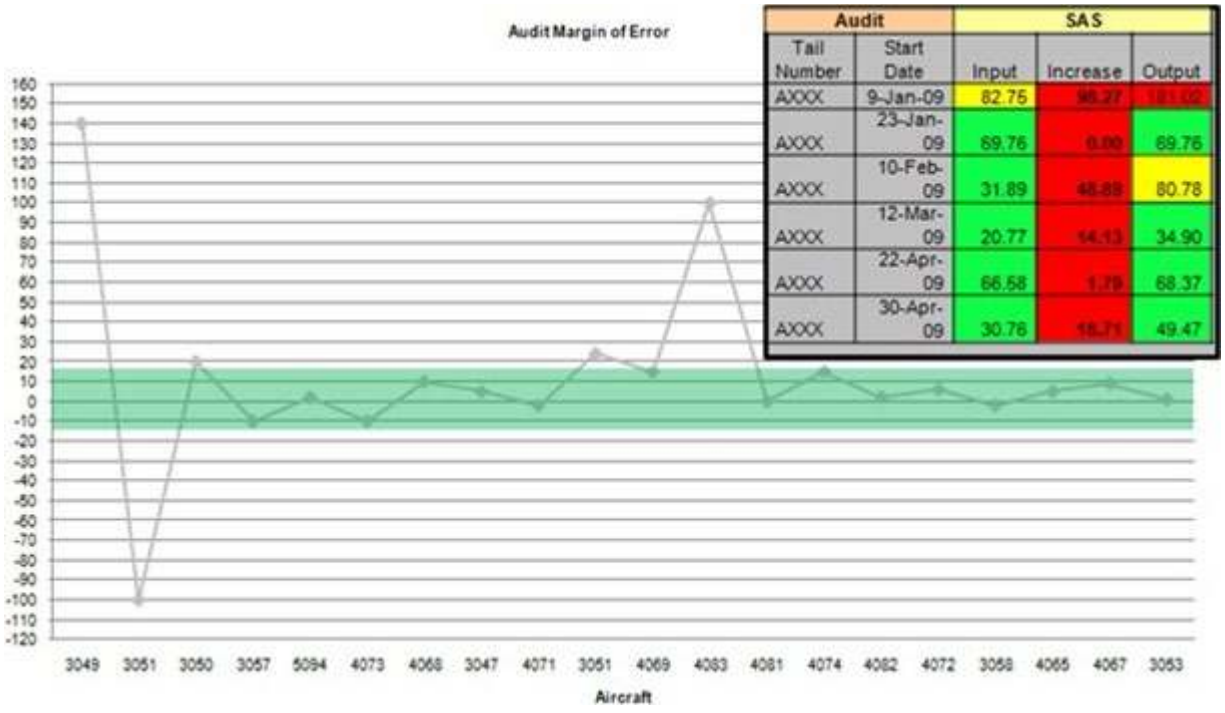


Attachment 3

F-22 OUTER MOLD LINE (OML) AUDIT

A3.1. F-22 Outer Mold Line (OML) Audit. The LO section is responsible to perform a SAS and aircraft OML audit on each assigned aircraft annually. **(T-3)** The audit is used to confirm that damage defects entered in the SAS during daily OML inspections, or damages removed from SAS during routine maintenance, provide for an accurate representation of the LO system health. Therefore, it must be completed by a separate entity within the LO section, excluding OML team members. **(T-3)** This can only be performed by physically matching aircraft damages with the entries in SAS. Errors identified during the audit will be recorded in the audit historical file and corrected in the SAS immediately. Any aircraft audit that results in a +15% or -15% SAS margin deviation indicates a potential deficiency with the OML inspection process. Maintenance supervisors with direct oversight of LO processes will ensure significant variances are understood and corrective actions (e.g., training, additional QA oversight, process changes) immediately follow. Internal root cause analysis and corrective actions must be documented for historical purposes any time an audit exceeds the margin percentages outlined above. **(T-3) Note:** There is an unacceptable risk to aircraft radar cross section and aircraft survivability due to substandard maintenance practices or inaccurate maintenance documentation into the SAS. Aircraft scheduled for an audit should be identified during the monthly/weekly shared resources meeting.

Figure A3.1. Sample Audit Metric.

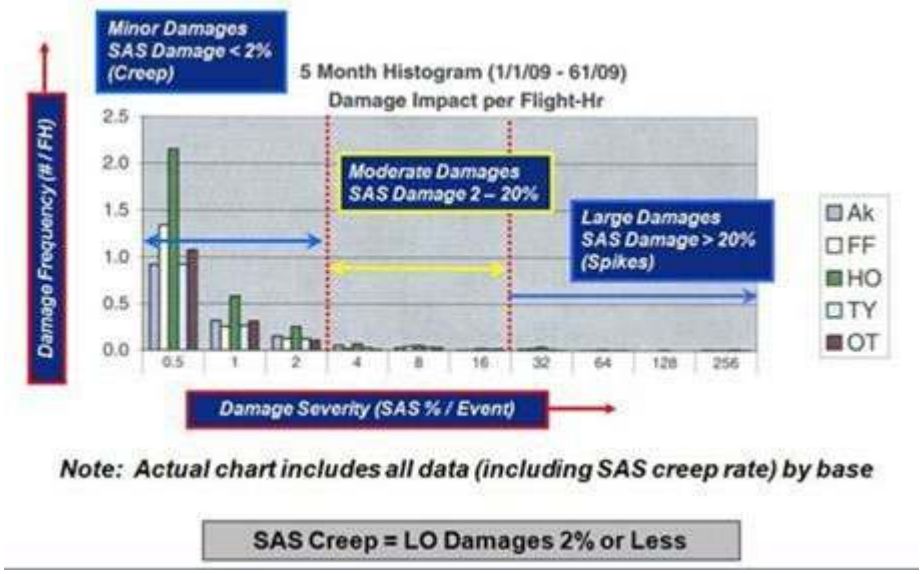


Audit		SAS		
Tail Number	Start Date	Input	Increase	Output
AXXX	9-Jan-09	82.75	98.27	181.02
AXXX	23-Jan-09	89.76	0.00	89.76
AXXX	10-Feb-09	31.89	48.89	80.78
AXXX	12-Mar-09	20.77	14.13	34.90
AXXX	22-Apr-09	66.58	1.79	68.37
AXXX	30-Apr-09	30.76	18.71	49.47

Attachment 4

DAMAGE TYPE DEFINITION

Figure A4.1. Damage Type Definition.



Attachment 5

SAMPLE LO REDUX FLOW PLAN

A5.1. Sample LO REDUX Flow Plan. The LO sections will establish a LO maintenance flow plan for each long lane maintenance action of five or more days. **(T-3)** The flow plan will help keep the aircraft maintenance schedule focused on the expected completion time.

Table A5.1. LO Maintenance Flow Plan.

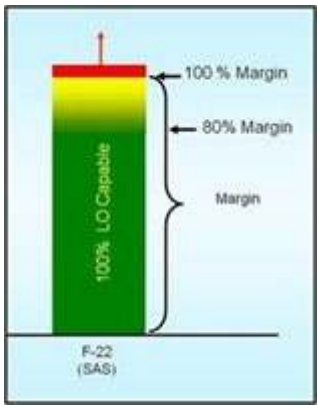
Calendar Days	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	
Production Day	1	2	3	4	5			6	7	8	9	10			11	12	13	14	15
Gap Filler Install FIP Casting																			
L04 Application / Top Primer Application																			
Boot / Butter / L07 TypeC / Flex primer																			
Topcoat																			
Touch Up / Panel Closure and Install																			

Attachment 6

SAS MARGIN DEFINITION

A6.1. SAS Margin Scale. The graphic below **(Figure A6.1)** depicts the SAS margin scale used to determine mission capable status for the F-22 LO system.

Figure A6.1. SAS Margin Definition.



A6.1.1. The LO maintenance margin for each aircraft is determined via an OML inspection and use of a SAS. An aircraft is fully mission capable for the LO system when the SAS margin is less than or equal to 100%. LO restoration is required when 100% SAS margin is exceeded. **(T-3)** Refer to the F-22 Mission Essential Systems List for LO status reporting.

A6.1.2. The desired fleet and/or aircraft signature margin is one that enables the unit to meet its DOC statement. Maintaining a fleet SAS margin average at or below 60% during peacetime operations will enable combat coded units to meet its DOC statement without requiring additional maintenance prior to deployment. Consider any individual aircraft that is at or exceeds 80% of its SAS margin as a priority for the next scheduled LO restoration.

A6.1.3. SAS does not rank order aircraft in terms of RCS.

A6.1.4. SAS number does not correlate to an RCS pattern.

A6.1.5. SAS is not a mission planning tool.

A6.1.6. OML inspections used to determine SAS margins may be accomplished daily or weekly by LO qualified signature management teams using procedures outlined in TOD. Combat coded units using a weekly OML inspection process must revert to a daily inspection during contingency operations. If a unit is using a weekly OML inspection, trained/qualified crew chiefs accomplish coating evaluations during post-flight inspections daily. The crew chief's role is to identify and report major LO discrepancies that pose a safety of flight risk, as to prevent propagation if unrepaired, and could jeopardize aircraft mission capable status for LO. Unit AFETS personnel are responsible for establishing and maintaining a robust APG training plan for weekly OML inspections. **(T-3)**

Attachment 7

DEBRIEF TAILORED CONSEQUENCES SUPPLEMENT

A7.1. Tailored Consequences. Table A7.1 represents tailored consequences in an incomplete format to match the classification of this document. This enables the reference of many tailored consequences during pilot debrief as outlined in Chapter 4, and as referenced by the F-22 MESL. Additionally, answers to Table 4.3 questions E and F can be informed by consulting this table. Table A7.1 may be interpreted as follows: an ICAW is listed with the individual tailored consequences beside it. Tailored consequences with an X in the INOP column are most likely to be assessed as non-operational per question E or generate an answer of NO per question F of Table 4.3 and are expected to drive a system capability Code 3 per Table 4.2. Tailored Consequences with an X in the DEGD column are most likely to be assessed as degraded per question E or generate an answer of YES per question F of Table 4.3 and are expected to drive a system capability Code 2 per Table 4.2.

Table A7.1. Tailored Consequences.

ICAW	Consequence	INOP	DEGD
AAI DEGD	ALL CONSEQUENCES		X
AAI FAIL	ALL CONSEQUENCES	X	
CNI RECONFIG	ALL PERSISTENT CNI RECONFIG	X	
CM DEGD	ALL CONSEQUENCES	X	
CM FAIL	ALL CONSEQUENCES	X	
EW DEGD	ALL CONSEQUENCES	X	
EW DOOR	ALL CONSEQUENCES	X	
EW FAIL	ALL CONSEQUENCES	X	
EXCM DEGD	ALL CONSEQUENCES	X	
EXCM DOORS	ALL CONSEQUENCES	X	
EXCM FAIL	ALL CONSEQUENCES	X	
MLD DEGD	UPPER FORWARD MLD INOP	X	
	LOWER FORMWARD MLD INOP	X	
	UPPER AFT MLD INOPERATIVE	X	
	LOWER AFT MLD INOPERATIVE	X	
	LEFT SIDE MLD INOPERATIVE	X	
	RIGHT SIDE MLD INOPERATIVE	X	
	UPPER FORWARD MLD DEGRADED		X
	LOWER FORWARD MLD DEGRADED		X
	UPPER AFT MLD DEGRADED		X
	LOWER AFT MLD DEGRADED		X
	RIGHT SIDE MLD DEGRADED		X
	LEFT SIDE MLD DEGRADED		X
MLD FAIL	ALL MLDS INOP	X	
	FAILURE OF MLD SOFTWARE IN THE CIP	X	
NBILST FAIL	NBILST ACQUISITION AND TRACKING INOP	X	
RW DEGD	FWD L RW/DF BAND 3 ACQUISITION AND TRACKING INOPERATIVE	X	
	FWD R RW/DF BAND 3 ACQUISITION AND TRACKING INOPERATIVE	X	
	FWD L RW/DF BAND 4 ACQUISITION AND TRACKING INOPERATIVE	X	

	FWD R RW/DF BAND 4 ACQUISITION AND TRACKING INOPERATIVE	X	
	AFT L RW/DF BAND 3 ACQUISITION AND TRACKING INOPERATIVE	X	
	AFT R RW/DF BAND 3 ACQUISITION AND TRACKING INOPERATIVE	X	
	AFT L RW/DF BAND 4 ACQUISITION AND TRACKING INOPERATIVE	X	
	AFT R RW/DF BAND 4 ACQUISITION AND TRACKING INOPERATIVE	X	
	FWD L ELEVATION RW/DF BAND 4 ACQUISITION AND DF INOP	X	
	FWD R ELEVATION RW/DF BAND 4 ACQUISITION AND DF INOP	X	
	ALL QUADS RW/DF BAND 3 AND 4 ACQUISITION AND TRACKING DEGD	X	
	FWD L RW/DF BAND 3 ACQUISITION AND TRACKING DEGRADED		X
	FWD R RW/DF BAND 3 ACQUISITION AND TRACKING DEGRADED		X
	FWD L RW/DF BAND 4 ACQUISITION AND TRACKING DEGRADED		X
	FWD R RW/DF BAND 4 ACQUISITION AND TRACKING DEGRADED		X
	AFT L RW/DF BAND 3 ACQUISITION AND TRACKING DEGRADED		X
	AFT R RW/DF BAND 3 ACQUISITION AND TRACKING DEGRADED		X
	AFT L RW/DF BAND 4 ACQUISITION AND TRACKING DEGRADED		X
	AFT R RW/DF BAND 4 ACQUISITION AND TRACKING DEGRADED		X
	FWD L ELEVATION RW/DF BAND 4 ACQUISITION AND DF DEGRADED		X
	FWD R ELEVATION RW/DF BAND 4 ACQUISITION AND DF DEGRADED		X
RW SYS DEGD	FWD R RWR AND NBILT INOPERATIVE	X	
	FWD L RWR AND NBISLT INOPERATIVE	X	
	RW/DF BAND 3 AND 4 ACQUISITION AND TRACKING INOPERATIVE	X	
	ALL QUADRANTS RW/DF BAND 3 AND 4 DEGRADED	X	
	FWD L RW/DF BAND 3 AND 4 DEGRADED	X	
	FWD R RW/DF BAND 3 AND 4 DEGRADED	X	
	AFT L RW/DF BAND 3 AND 4 DEGRADED	X	
	AFT R RW/DF BAND 3 AND 4 DEGRADED	X	
	AFT L RW/DF AND NBILST INCREASED FALSE ALARM RATE		X
	AFT R RW/DF AND NBILST INCREASED FALSE ALARM RATE		X
	NBILST ACQUISITION AND TRACKING DEGRADED	X	
RW SYS FAIL	RW SYSTEM INOPERATIVE	X	
	RW SYSTEM LOST COMMUNICATION WITH THE CIPS	X	
RADAR DEGD	NO LONG-RANGE SEARCH	X	
	NO VELOCITY SEARC/ RANGE SEARCH	X	
	DEGRADED VELOCITY RANGE SEARCH	X	
	DEGRADED RANGE AND TIMLEINE PERFORMANCE IN SEARCH AND TRACK	X	
	DEGRADED LOOK DOWN AND SEARCH AND TRACK	X	

	DEGRADED LOW ALTITUDE SERCH AND TRACK	X	
	DEGRADED TIMELINE	X	
	DEGRADED AUTO ACQ/WAT	X	
	DEGRADED ANGLE ACCURACY IN SEARCH AND TRACK	X	
	DEGRADED RANGE PERFORMANCE IN ALL MODES	X	
	NO SAR MAPS		X
	DEGRADED SAR MAPS		X
	NO WEATHER MAP		X
	DEGRADED WEATHER MAP		X
	DEGRADED MISSILE QUALITY TRACK RANGE	X	
	NO MISSILE UPDATE	X	
	NO MISSILE DETECT	X	
	NO RAID ASSESSMENT		X
	NO NCTR	X	
	NO ELECTRONIC ATTACK		X
	DEGRADED ELECTRONIC ATTACK	X	
	DEGRADED ELECTRONIC PROTECTION	X	
	DEGRADED SAR MAP (GENERATION RECALL OR STORAGE)		X
	++FALSE TARGETS	X	
	++SYSTEM DEGRADATION	X	
	DEGRADED SEARCH AND TRACK	X	
	DEGRADED MISSILE DOWNLINK	X	
	++ NO ELECTRONIC ATTACK		
RADAR FAIL	NO RADAR	X	
	NO SAR MAPS	X	
	DEGRADED EP	X	
	DEGRADED RADAR INTERLEAVING	X	
	MISSILE BORESIGHT LAUNCH ONLY	X	
	NO RADAR EXCEPT MISSILE UPLINK	X	